SGA 2: Local Cohomology and Lefschetz Theorems

Exposé VIII: The Finiteness Theorem

Bounded source-aligned working unit — 19 July 2026

Authority. Corrected French lines 2553–2559; original printed p. 85; physical source-PDF p. 76; recomposed running p. 68. This bounded unit is translated and independently source-reviewed against the French TeX and direct PDF. Cumulative integration remains pending. Line 2560 is blank; the proof begins at line 2561 and is excluded.

It remains for us to prove the following result.

Lemma 1.2. — Let A be a Noetherian ring and let $\mathcal{C}$ be the category of A-modules. Let $\underline{\mathrm{F}}\colon \mathcal{C} \longrightarrow \mathbf{Ab}$ be a left exact additive functor such that, for every finitely generated A-module N and every injective A-module C, one has

$$\mathrm{R}^1\underline{\mathrm{F}}(\mathrm{Hom}(N, C)) = 0.$$

Let M be a finitely generated A-module of finite projective dimension. There is a spectral sequence

$$\mathrm{R}^*\underline{\mathrm{F}}(M) \Longleftarrow \mathrm{Ext}^p_{\underline{\mathrm{F}}}(\mathrm{Ext}^{-q}(M, A), A),$$

where $\mathrm{Ext}^p_{\underline{\mathrm{F}}}$ denotes the p-th right derived functor of $\underline{\mathrm{F}} \circ \mathrm{Hom}$.